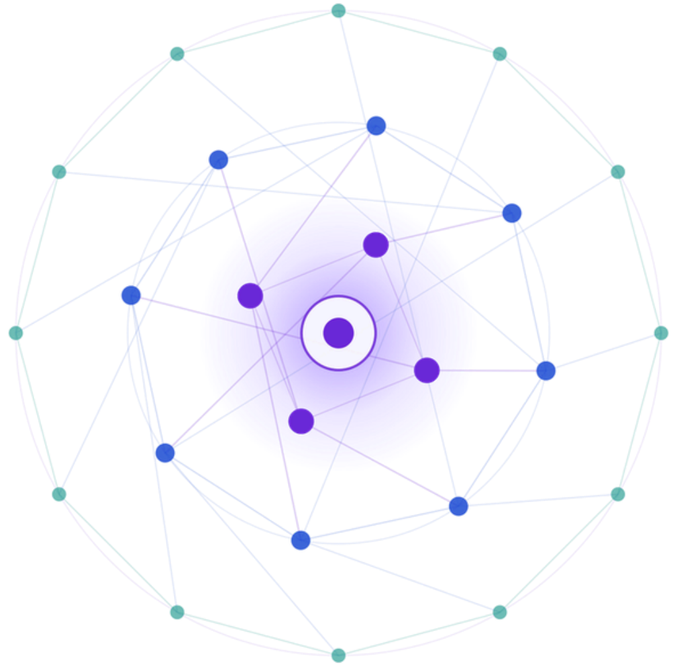
SAMSUNG PRISM

Generative AI Hackathon



3rd Edition 2026 – 27

Theme ID: **Theme 2: Smart Guided Troubleshooting Engine**

Team Name: **Code Brewers**

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Submission Github link: **github.com/Bhavy-official/Galaxy-Resolve**

Theme

Theme 2 · Smart Guided Troubleshooting Engine

Turn vague Galaxy device complaints into grounded, ordered troubleshooting plans with verified One UI deeplinks.

The problem today

Users describe issues loosely: “my battery drains fast”, “my screen is flickering”.

Support relies on SIIS articles
Agents manually interpret complaints and select troubleshooting steps.

Customers navigate Settings manually:
e.g. Settings → Display → Navigation bar.

Key gaps:
Manual interpretation is slow, Settings navigation is friction-heavy, and unconstrained GenAI can hallucinate unsupported actions.

What the engine must do

1. **Ground the complaint**
   Extract intent and actions only from verified SIIS source content.
2. **Resolve the exact screen**
   Map each action to a verified One UI deeplink from the 578-entry catalog.
3. **Sequence safely & respond fast**
   Safe actions first, critical operations last; repeated queries are served through the fast cache.

Vague complaint + SIIS article → **Galaxy-Resolve Engine** → **Grounded, Schema-Valid Plan + One-Tap Deeplinks**

Existing Solutions & Gaps

Existing Approach	How it works	Key gap
Manual Agent triage	Agents interpret user complaints using internal SIIS knowledge.	Slow, costly, and remediation can vary between agents.
Static FAQs & Search	Users search support articles for troubleshooting instructions.	Long instructions and multi-level One UI navigation create high friction.
Generic LLM / RAG	LLM generates conversational troubleshooting steps.	Can hallucinate settings paths, deeplinks, and unsafe recommendations.
Keyword-Only Retrieval	Matches complaint words against available support/settings entries.	Struggles with paraphrased complaints and semantic intent.
Unverified Deeplink Selection	Model directly chooses a settings link from available options.	May select broad parent menus instead of the exact actionable screen.

THE GAP WE ADDRESS

Galaxy-Resolve bridges the gap between understanding a user's complaint and actually resolving it by combining grounded troubleshooting, verified One UI deeplinks, safety-first sequencing and deterministic validation into a single actionable, one-tap experience.

Galaxy-Resolve: Understand the complaint → Verify the action → Open the exact screen → Resolve safely in one tap

Our Solutions & Architecture Diagram

1 GROUNDED INTELLIGENCE

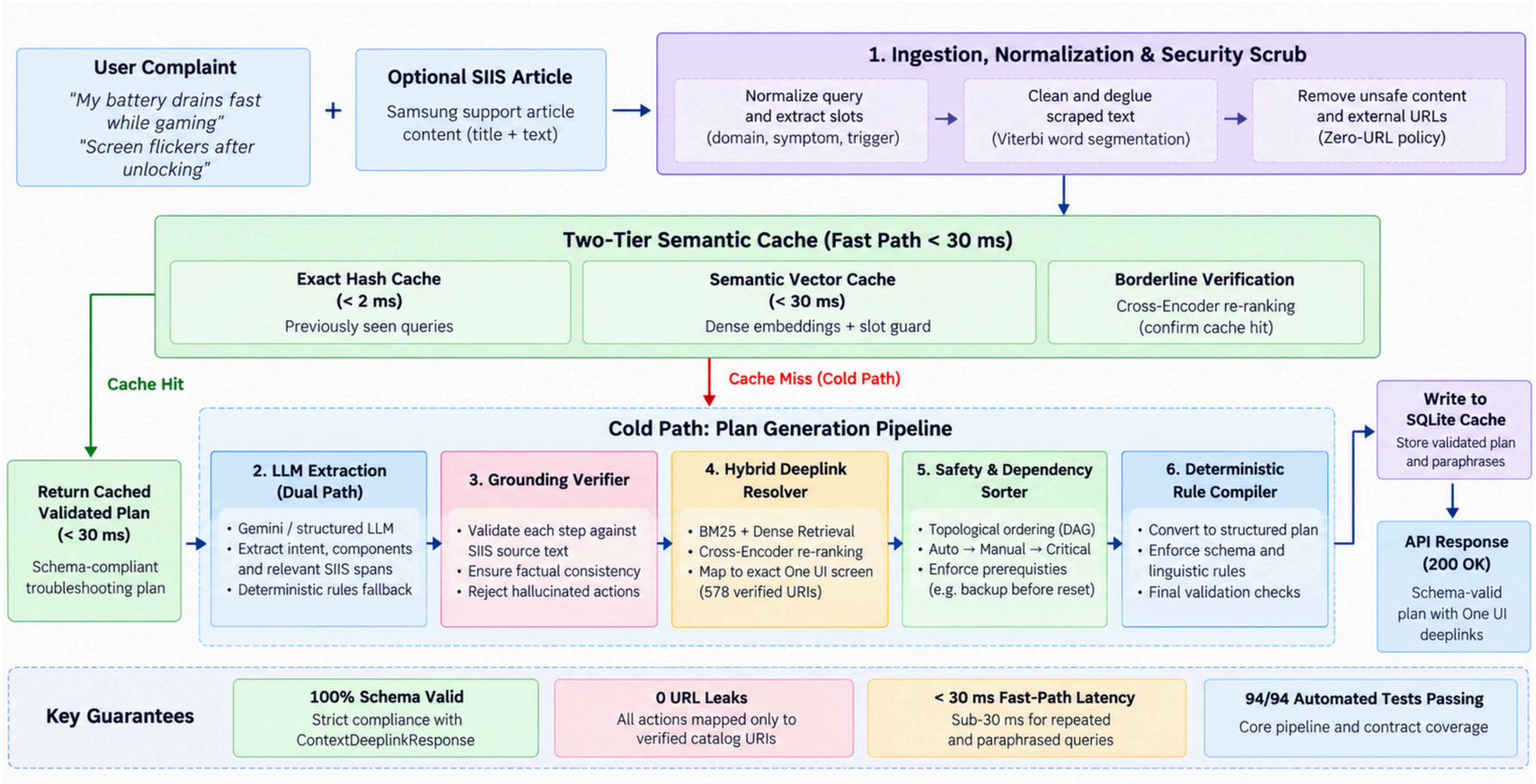
LLM extracts user intent and relevant SIIS evidence, while deterministic code prevents unsupported actions.

2 VERIFIED ONE UI ACTIONS

Hybrid retrieval (BM25 + Dense Embeddings + Cross-Encoder) maps each action to the exact verified deeplink from the 578-entry catalog.

3 SAFE & FAST EXECUTION

Safety rules prioritize non-invasive actions before critical operations, while the semantic cache enables sub-30 ms fast-path responses.



CORE PRINCIPLE: LLM Proposes → Code Verifies → One UI Executes

Demo & Product Walkthrough

1. **Complaint + SIIS context**
   Enter a natural-language Galaxy complaint such as “My phone screen flickers after unlocking” with the relevant Samsung Support article.

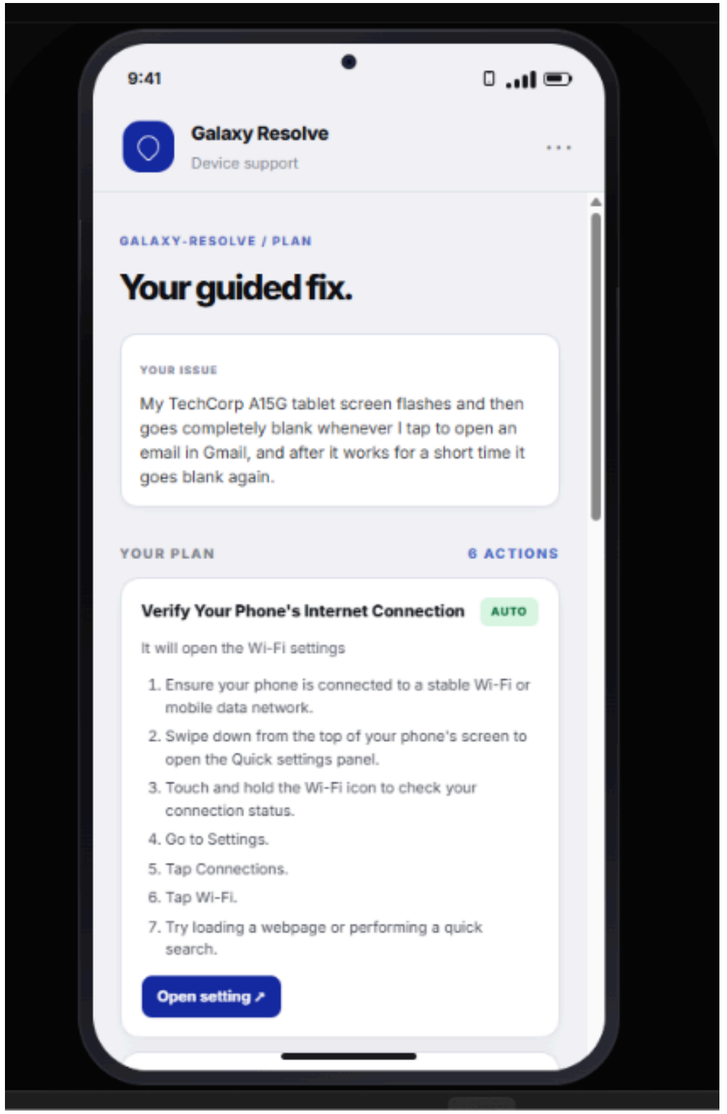
2. **Understand the issue**
   The engine normalizes the complaint and extracts the intent, relevant components, symptoms, and supported troubleshooting actions.

3. **Prove the action is grounded**
   Each generated step is checked against the SIIS source text.
   Unsupported or hallucinated actions are rejected.

4. **Open the exact screen**
   Hybrid retrieval maps the action to a verified One UI deeplink from the 578-entry catalog, enabling direct Settings navigation.

5. **Ask again, respond faster**
   Repeated or paraphrased complaints can be served through the semantic cache in under 30 ms, while maintaining slot and similarity checks.

FROM COMPLAINT TO ACTION:
Understand the issue → Ground the solution → Open the exact setting → Resolve



(Image from actual project’s frontend)

Tools and tech stack used

1. GenAI & Extraction

- Google Gemini — Structured LLM extraction
- Intent, component & SIIS span extraction
- Deterministic rules fallback
- Structured JSON outputs

2. Retrieval & Deeplink Mapping

- all-MiniLM-L6-v2 — Dense embeddings
- BM25 — Keyword retrieval
- Cross-Encoder reranking
- Reciprocal Rank Fusion (RRF)
- 578 verified One UI deeplinks

3. Backend & API

- Python 3.10+
- FastAPI + Uvicorn
- Pydantic — Schema validation
- REST API + SSE streaming
- Real-time request tracing & metrics

4. Cache & Storage

- SQLite — Persistent cache & traces
- all-MiniLM-L6-v2 embeddings
- Two-tier semantic cache
- Deterministic slot guards
- Precomputed troubleshooting plans

5. Security & Safety

- Zero-URL security scrubber
- SIIS grounding verification
- Viterbi word degluing
- Topological dependency ordering
- Auto → Manual → Critical action hierarchy

6. UI & Deployment

- React based frontend
- Embedded Galaxy One UI visualizer
- Live SSE execution trace
- Simulated voiceassist:// deeplink actions
- Single FastAPI service

Impact & Use case

USER EXPERIENCE

One-Tap Troubleshooting

- Understands natural-language complaints
- Grounds actions in Samsung Support content
- Opens the exact verified One UI screen
- Avoids multi-level Settings navigation

SUPPORT OPERATIONS

Faster & More Consistent Support

- Grounds troubleshooting in SIIS content
- Reduces repetitive manual interpretation
- Produces consistent, schema-valid plans
- Enforces safe action ordering

GALAXY ECOSYSTEM

Multiple Troubleshooting Entry Points

- Samsung Support content → actionable troubleshooting
- Galaxy Settings - verified deeplink actions
- Voice-driven / conversational input - troubleshooting plan
- Reusable API - integration-ready engine

<30 ms	578	100%	0
Fast-path response	**Verified One UI deeplinks**	**Schema-valid output**	**External URL leaks**

KEY IMPACT: Less navigation • Grounded answers • Safer action ordering • Faster resolution

Innovation highlights, results and limitations

What makes Galaxy Resolve different?

LLM Proposes → Code Disposes

Galaxy Resolve combines semantic intelligence with deterministic verification to deliver grounded, precise, safe and actionable troubleshooting.

What we achieved

100%	578	94/94	<30 ms	0
schema-valid	Verified One UI deeplinks	Tests Passing	Fast-Path Latency	URL leaks

Core innovations

Grounded extraction
Troubleshooting actions are derived from verified SIIS evidence.

Hybrid screen resolution
BM25 + dense retrieval + Cross-Encoder identify the exact actionable One UI screen.

Safety-aware execution
Dependency graphs and severity ordering ensure safe actions precede disruptive operations.

Mismatch protection
Unsupported physical actions require safe manual intervention.

Current boundaries

Catalog coverage
Limited to the verified 578-entry deeplink catalog.

SIIS dependency
No source evidence → no invented troubleshooting step.

Device & version dependence
Different One UI versions may require separate catalog profiles.

Hardware actions
Unsupported physical actions require safe manual intervention.

What’s next

01 Validate & Expand

Verify on real Galaxy devices

Test resolved actions directly on Galaxy devices and confirm each deeplink reaches the intended screen.

Expand the URI catalog

Add missing hardware/system actions and device- and One UI-version-specific deeplink profiles.

02 Move Intelligence On-Device

Private edge intelligence

Move extraction and retrieval toward Gemini Nano / ONNX + Galaxy NPU for low-latency, privacy-preserving troubleshooting.

Offline-capable troubleshooting

Reduce dependence on cloud inference by bringing more of the troubleshooting pipeline onto the device.

03 Close the Loop & Scale

Outcome-aware troubleshooting

Learn from Resolved / Not Resolved feedback and dynamically select the next best troubleshooting step.

Cross-device Galaxy support

Extend troubleshooting to Galaxy Watch, Buds and SmartThings, including multimodal inputs such as photos and screen recordings.

Validate on-device → Move intelligence closer to the device → Learn from outcomes → Scale across Galaxy

From guided troubleshooting to proactive, outcome-aware Galaxy intelligence.

Brownie points (differentiation)

1 Intelligent Control

LLM handles semantic extraction; deterministic code controls grounding, deeplinks, schema and safety rules.

2 Exact Screen Mapping

BM25 + Dense Retrieval + Cross-Encoder resolve the exact actionable One UI screen from the verified catalog.

3 Evidence-backed actions

Every troubleshooting step is verified against SIIS source content; unsupported actions are rejected.

4 Secure URL handling

Deterministic URL scrubbing prevents external or fabricated links from reaching the final response.

5 Safe Action Sequencing

Topological dependencies and severity rules ensure safe actions come first and disruptive operations come last.

6 Fast, Reliable Responses

Two-tier semantic caching enables sub-30 ms fast-path responses with strict schema validation.

Checklist: updated on public GitHub

- Working prototype code, public GitHub repo: **Yes**
  github.com/Bhavy-official/Galaxy-Resolve
- README with reproducible setup instructions: **Yes**
- Demo video, max 5 minutes: **Yes**
- Presentation file (PPT or PDF): **Yes**

Thank you

Team Code Brewers · SRM

Organised by the Language AI Team and the PRISM Team, Samsung R&D Institute India